

Basant Hesham

Software Engineer

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Summary

A skilled software engineering student with a background in web development. Proficient in creating innovative solutions using Java, C/C++, JavaScript, React, and HTML. Experienced in collaborating with teams and delivering high-quality projects and responsive web applications. Committed to continuous learning and personal development.

Education

BEng Software Engineering

York University

Undergraduate student at Lassonde School of Engineering

DEC 2026

Certifications

Meta Front-End Developer (access certificate [here](#))

Sep 2024

Programming with JavaScript by Meta (access certificate [here](#))

July 2024

HTML and CSS in depth by Meta (access certificate [here](#))

Aug 2024

Technical Skills

Programming Languages: Java, JavaScript, C/C++, Python, SQL, Bash

Web Technologies: HTML, CSS, React

Tools & Platforms: IntelliJ IDEA, Visual Studio Code, Git, Linux, MATLAB

Other Skills: Data Analysis, Problem-Solving, Team Collaboration

Experience

Computer Lab Assistant

York University, Toronto, ON- SEP 2022 – APR 2023

- Managed laboratory equipment configurations, including hardware and software such as lab-test, operating systems like Linux and Windows.
- Provided technical support and guidance to over 100 students daily, assisting with lab experiments and troubleshooting issues.
- Supported researchers with technical tasks, equipment troubleshooting, and experiment setup.
- Conducted routine equipment maintenance and calibration to minimize downtime and enhance reliability.
- Oversaw daily laboratory operations, ensuring compliance with safety standards, cleanliness, and proper resource allocation.
- Applied expertise in Linux system administration, shell scripting, MATLAB, Java, and C to manage operating systems and networks efficiently.

Projects:

Personal portfolio

Toronto, ON- DEC 2024

- Personal [portfolio](#), GitHub [link](#).
- Designed and developed a personal portfolio website using React, HTML, and JavaScript.
- Built with a responsive design, ensuring compatibility across both desktop and mobile devices.
- Implemented a toggle feature for dark and light themes, allowing users to choose their preferred viewing mode.
- Utilized Visual Studio Code (VSC) for efficient code writing and project management.
- Created an interactive and user-friendly layout to showcase skills, projects, and experience.
- Applied modern web development practices for optimal performance and cross-browser compatibility.

Solution for Booking a Table on the Little Lemon Website

SEP.2024

Coursera - Meta Front-End Developer Capstone Project: [GitHub link](#)

- Developed a web application to facilitate table bookings on the Little Lemon restaurant website as part of the Meta Front-End Developer Capstone project.
- Designed the user interface (UI) to be intuitive, allowing customers to select booking details such as date, time, and number of guests.
- Implemented responsive design techniques to ensure the website is functional on both desktop and mobile devices.
- Integrated JavaScript for dynamic user interactions and validation of booking information.
- Applied front-end development principles such as user experience (UX) optimization, efficient navigation, and accessibility features.
- Collaborated with real-world best practices in web development, utilizing tools like Git and GitHub for version control.

Digital Filters and Noise Removal from Audio Signals

York University -Toronto, ON - DEC 2024

- Used MATLAB to write code for designing a bandpass FIR filter and removing noise from audio signals.
- Implemented FFT to analyze the noisy signal, identify noise frequencies, and applied time-domain and frequency-domain filtering techniques.
- Created MATLAB plots to visualize the frequency spectrum, filter performance, and cleaned audio signals.
- Validated the filter performance by analyzing time-domain and frequency-domain responses through plots.
- Access code document from [here](#).

Database System Project

York University -Toronto, ON- MAY 2023 - AUG 2023

- Built a Java-based GUI for data input and displaying MySQL query results.
- Integrated methods to streamline data insertion, retrieval, and backend processes.
- Optimized database performance using SQL Trace, Explain Plan, and SQL Loader.
- Troubleshoot and improved database functionality and readability.
- Gained experience in ETL processes, data conversion, and data analysis.

Auto plant watering system

York University -Toronto, ON- JAN 2021 – APR 2021

- You can view the project in action through the following video: [Presentation Video](#), [GitHub link](#)
- Designed and built an automated plant watering system using Arduino microcontrollers and sensors.
- Programmed the system to monitor soil moisture and activate a pump when levels dropped below a set threshold.
- Integrated temperature and humidity sensors to prevent overwatering and optimize plant growth.
- Conducted thorough testing and troubleshooting to ensure system accuracy and reliability.

Arduino-MATLAB Temperature Alert System

Personal project -Toronto, ON- DEC 2021

- You can view the project in action through the following video: [Presentation Video](#)
- Developed a temperature alert system using an Arduino-compatible kit and MATLAB.
- Programmed MATLAB to detect temperature changes and process signals from the Arduino Grove board.
- Configured the Arduino board to trigger an audible alarm, notifying users to manage temperature levels.

Volunteer experience

ACSA Food Bank

Toronto, ON- OCT2020- NOV 2021

- Managed food donations and inventory, ensuring proper tracking and storage.
- Maintained facility cleanliness and adhered to safety standards.
- Assisted participants during food distribution events, ensuring smooth operations and addressing inquiries.
- Collaborated with team members to improve efficiency and enhance the experience for attendees.